Clickbait Headline Classification Using NLP and Machine Learning

Short Project Report based on Clickbait_Classification.ipynb

1. Project Overview

This project develops a natural language processing (NLP) classification system for detecting whether a news or media headline is clickbait. The notebook performs text exploration and preprocessing, including tokenization, lowercasing, stop-word removal, lemmatization, POS tagging, chunking, and named-entity recognition. It then converts headline text into TF-IDF features and trains machine-learning classifiers, including Logistic Regression and Linear SVM.

2. Dataset

Dataset file: clickbait_data.csv. The notebook reports 32,000 records and two original columns: **headline** and **clickbait**. The label distribution is almost perfectly balanced: 16,001 non-clickbait examples (0) and 15,999 clickbait examples (1).

The notebook does not document an external dataset source or license, so this report does not assign a specific public-dataset provenance beyond the filename used by the notebook.

3. Text Processing

The workflow includes word and sentence tokenization, lowercasing, removal of stop words, lemmatization with POS information, POS tagging, simple syntactic chunking, and NER experimentation. The final modeling workflow uses a cleaned/lemmatized text representation and TF-IDF vectorization.

4. Model Development

The notebook uses an 80/20 train-test split with stratification in the main modeling pipeline, producing 25,600 training examples and 6,400 test examples. TF-IDF is used to represent the text, with a maximum of 5,000 features in the final Linear SVM workflow.

5. Reported Results

Model / Run	Accuracy	Precision	Recall	F1
Logistic Regression + TF-IDF	96.34%	97.74%	95.05%	96.38%
Linear SVM + TF-IDF	97.17%	97.72%	96.73%	97.22%

Among the reported runs, Linear SVM achieved the strongest test performance with 97.17% accuracy and an F1 score of 97.22%. The notebook also includes a Gradio application based on a Linear SVM trained with TF-IDF features.

6. Interactive Application

The notebook contains a Gradio interface titled **Clickbait Headline Detector**. A user enters a headline and the trained Linear SVM returns either “Clickbait Headline” or “Not Clickbait.” The notebook used a temporary Gradio share URL during execution; such URLs are temporary and should not be treated as permanent deployment links.

7. Limitations

The notebook does not document the authoritative source/license of clickbait_data.csv. In addition, the notebook contains several exploratory NLP operations that are not all used as model features. For a production-quality implementation, preprocessing and vectorization should be placed in a reproducible pipeline and evaluated with cross-validation. A probability-calibrated model or decision score can also be added if confidence estimates are required.

8. Future Improvements

Possible improvements include hyperparameter tuning, cross-validation, stronger duplicate/leakage checks, comparison with Naive Bayes and Random Forest, ROC-AUC/PR-AUC analysis, model persistence with Joblib, and deployment as a permanent web application.

9. Conclusion

The project demonstrates a complete NLP classification workflow for clickbait detection, from text preprocessing and linguistic analysis to TF-IDF feature extraction, machine-learning classification, evaluation, and an interactive Gradio interface. Based on the reported notebook results, Linear SVM provides the best performance among the explicitly reported final model runs.